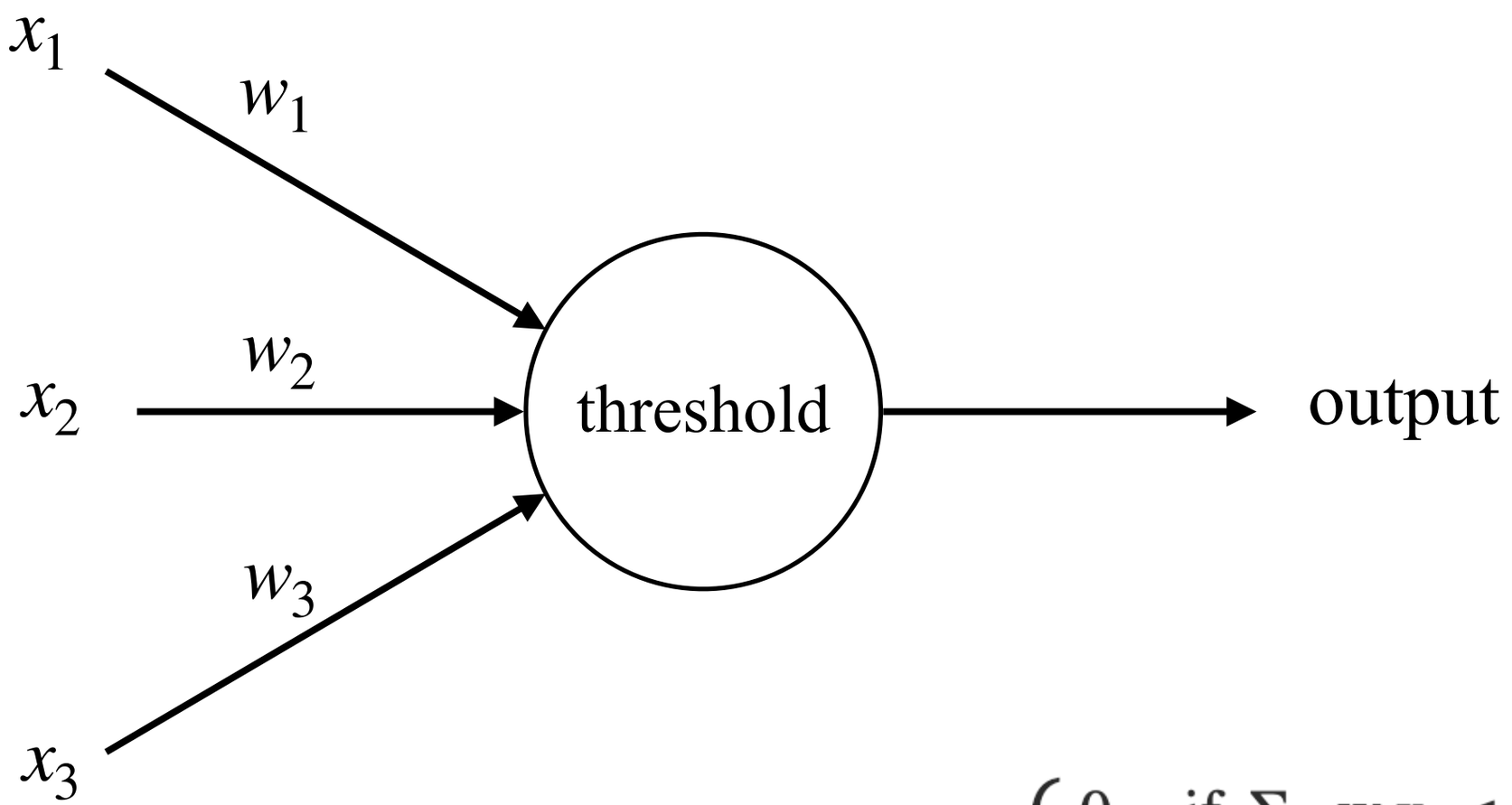
Neural networks

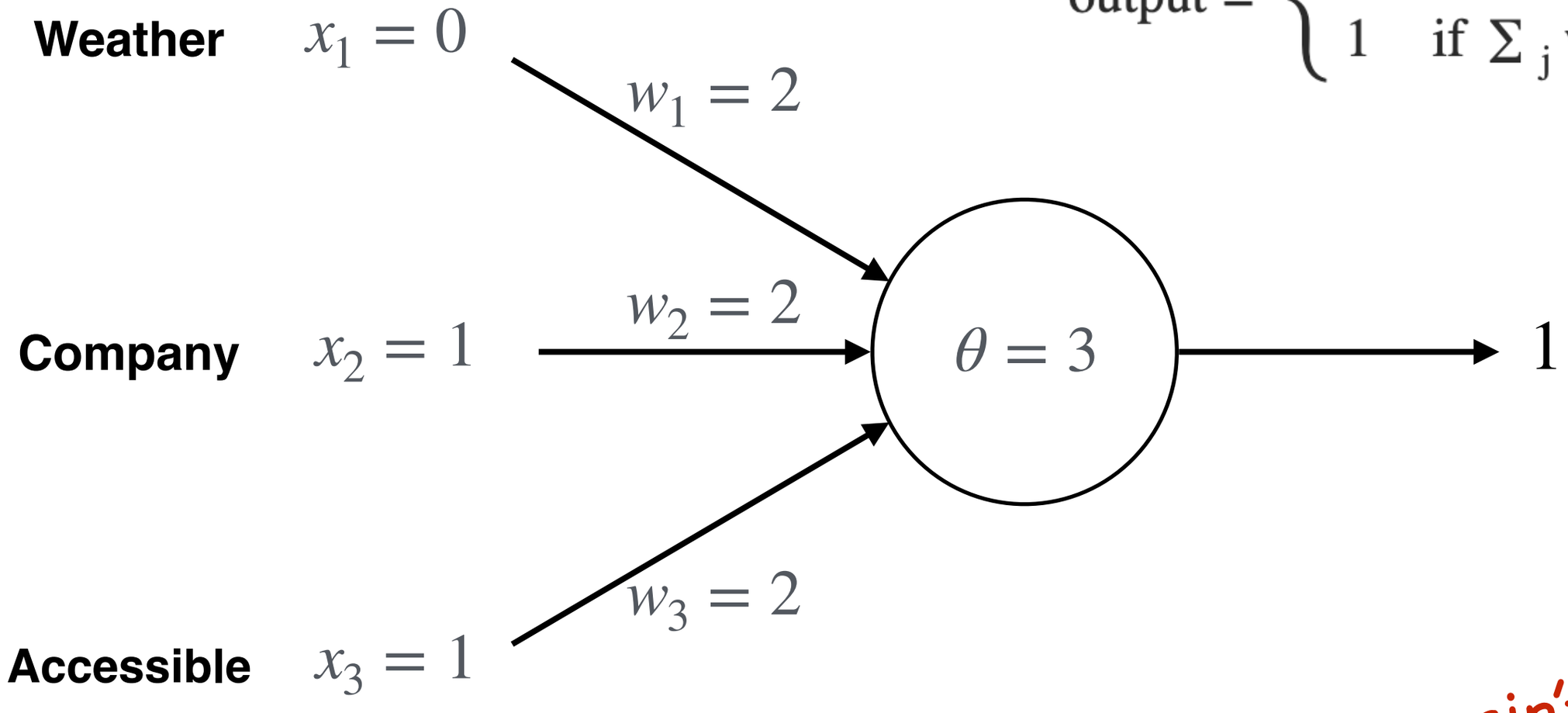
Perceptron



$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{threshold} \end{cases}$$

Perceptron

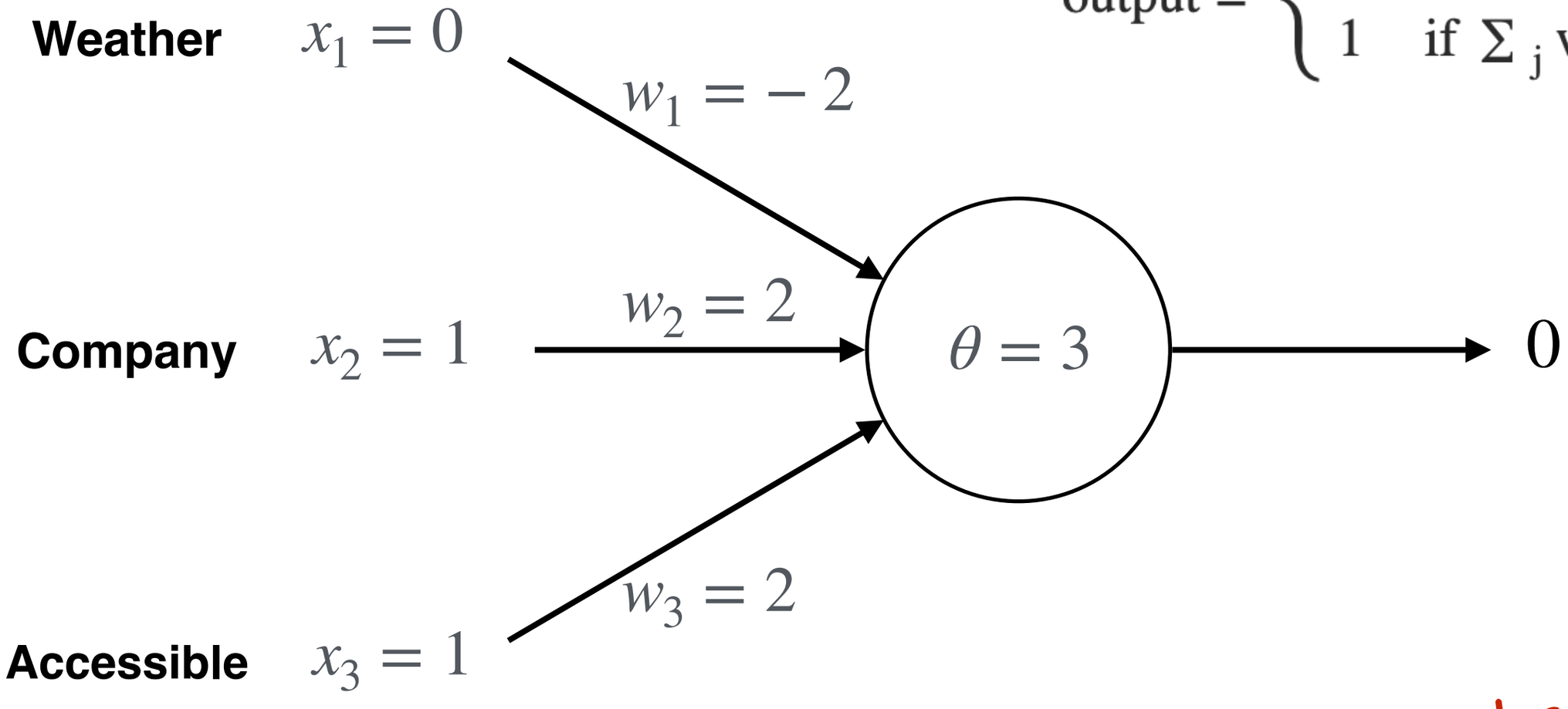
$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$



Two out of three ain't bad

Perceptron

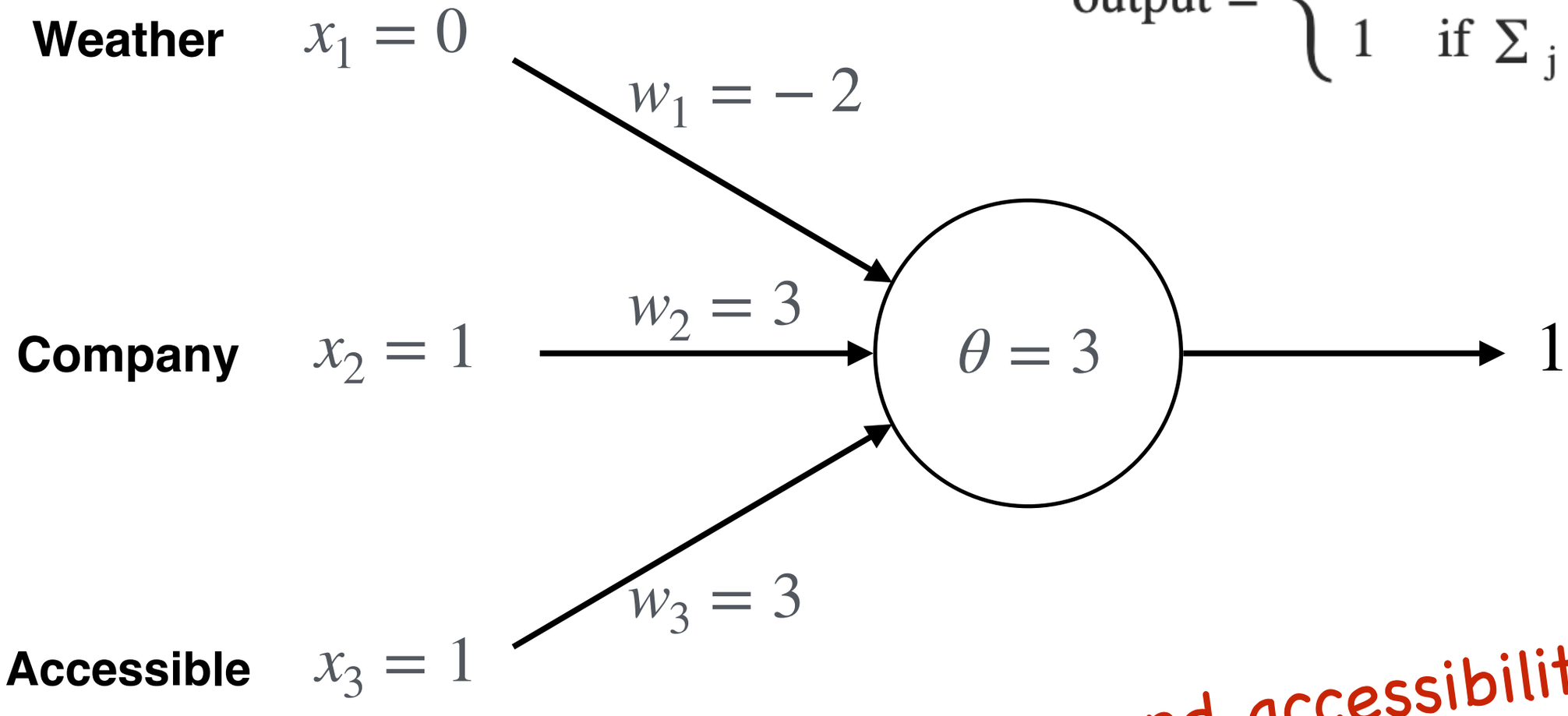
$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$



Not going anywhere in bad weather!

Perceptron

$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$



Company and accessibility makes up for bad weather

Bias (- threshold)

$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{threshold} \end{cases}$$

$$\sum_j w_j x_j \qquad b \equiv -\text{threshold}$$

$$\text{output} = \begin{cases} 0 & \text{if } w \cdot x + b \leq 0 \\ 1 & \text{if } w \cdot x + b > 0 \end{cases}$$

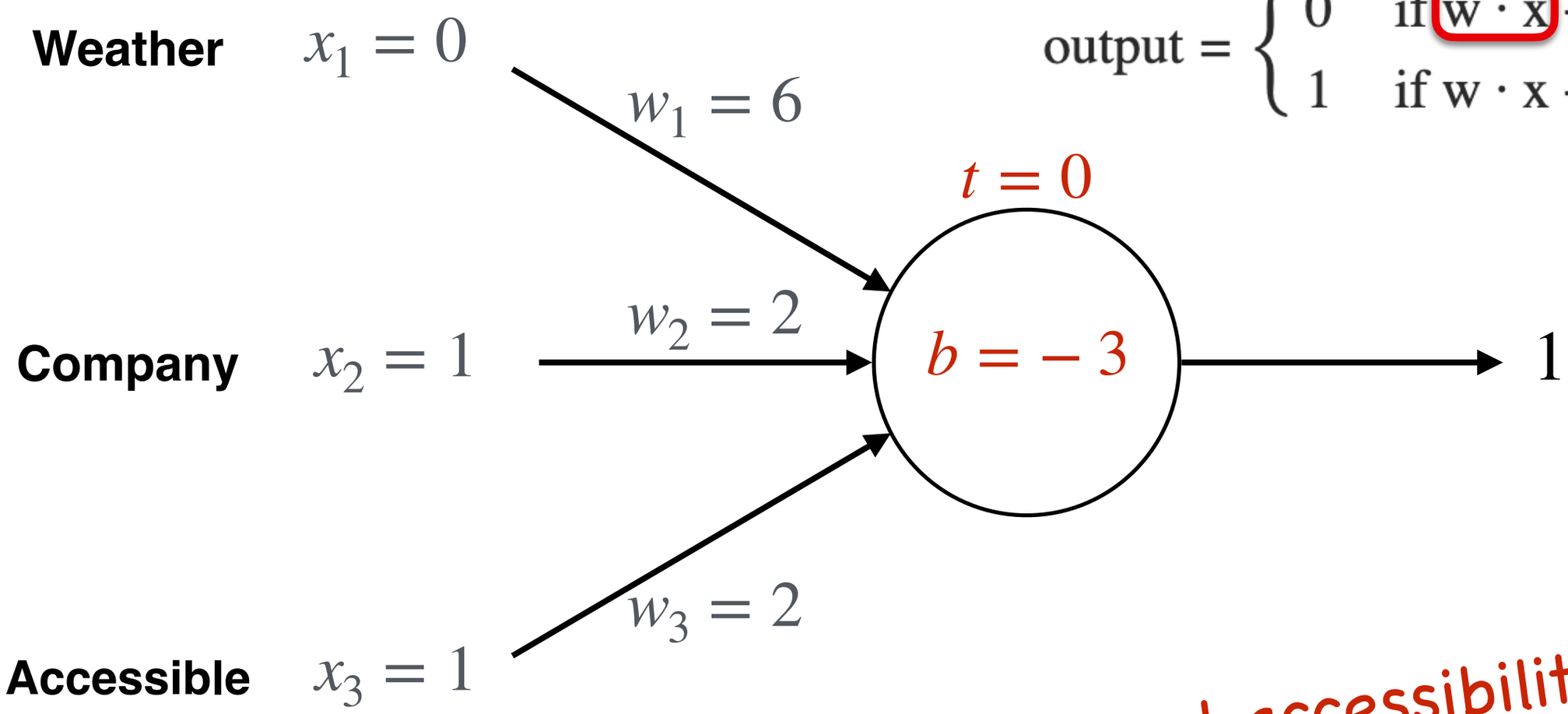
Bias > 0: outputs even with no input

Bias < 0: may not outputs even with some input

Perceptron

$$\text{output} = \begin{cases} 0 & \text{if } w \cdot x + b \leq 0 \\ 1 & \text{if } w \cdot x + b > 0 \end{cases}$$

$\sum_j w_j x_j$ $\qquad$ $b \equiv -\text{threshold}$

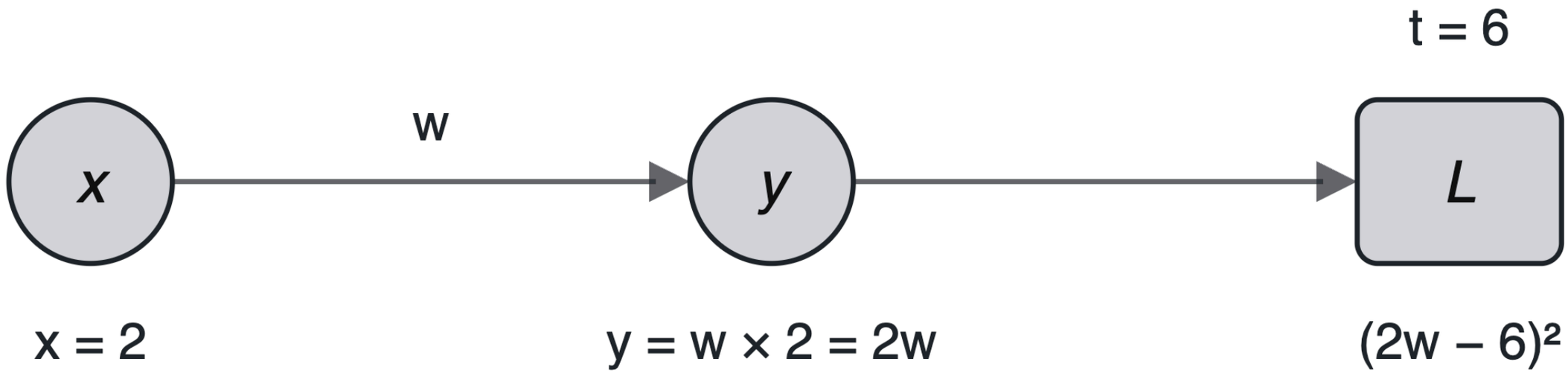


Company and accessibility makes up for bad weather

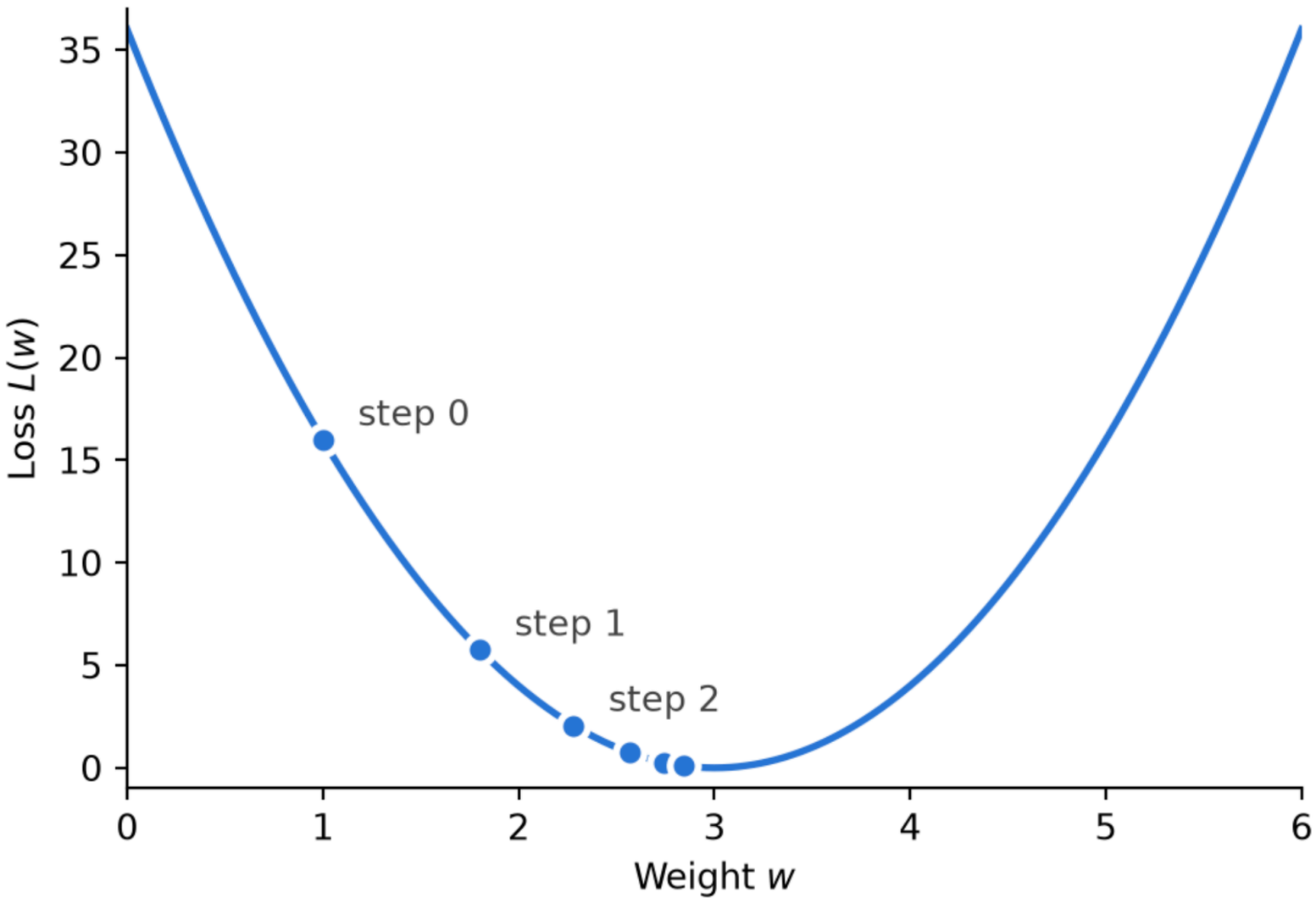
Training

A small change in weights or thresholds should cause a small change in the output (in the right direction).

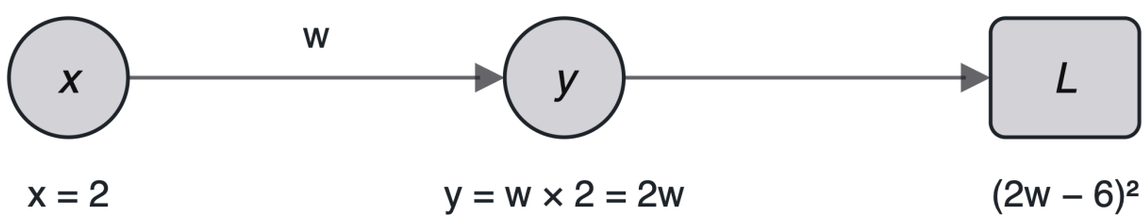
$$L(w) = (y - t)^2 = (2w - 6)^2$$

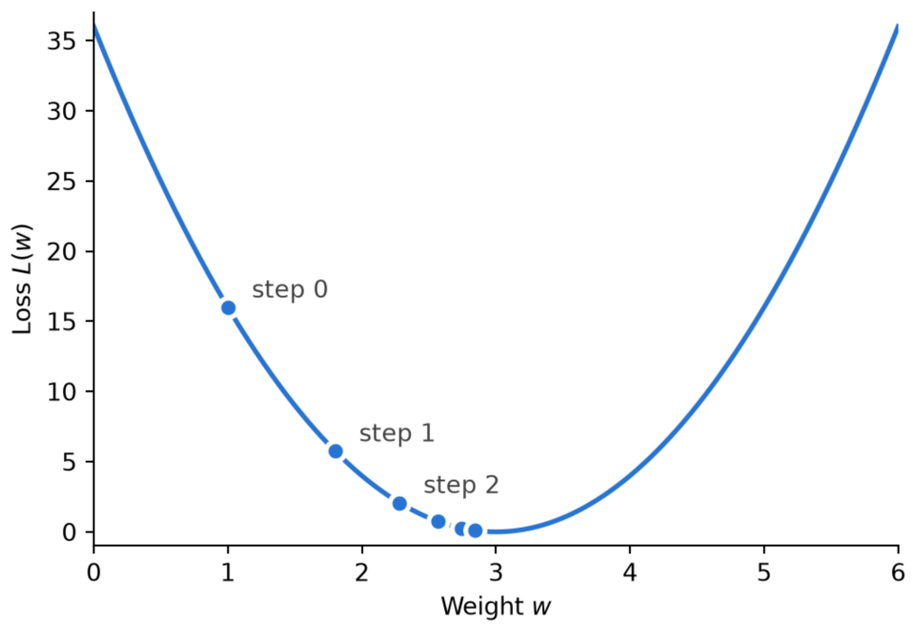


Training



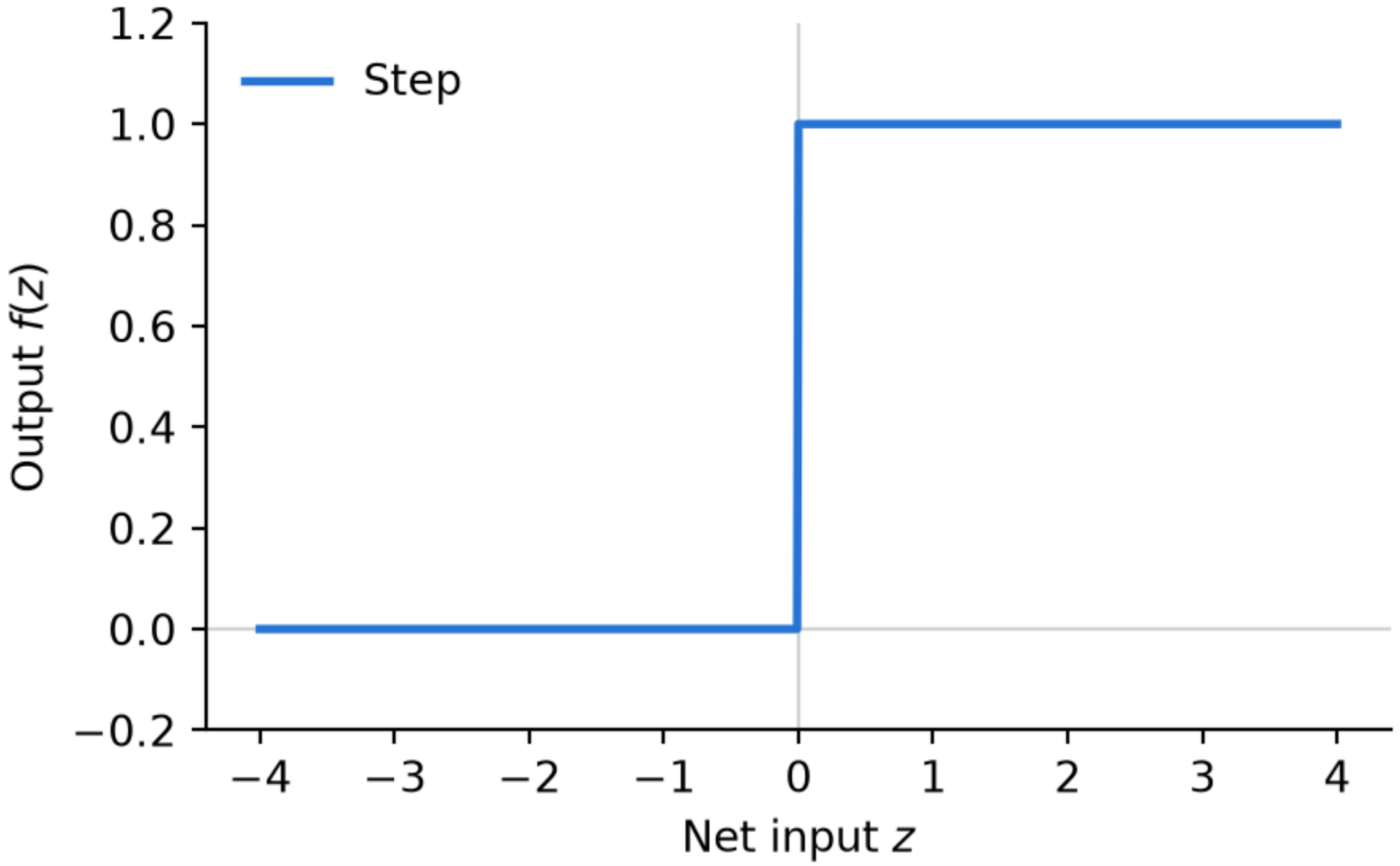
```
x = 2                                   # training input
target = 6                              # training output
w = 1                                   # starting guess
learning_rate = 0.05                    # step multiplier
for step in range(15):
    y = w * x                           # run the network
    loss = (y - target) ** 2            # how wrong is it?
    slope = 2 * (y - target) * x        # which direction?
    w = w - learning_rate * slope # take a step
```





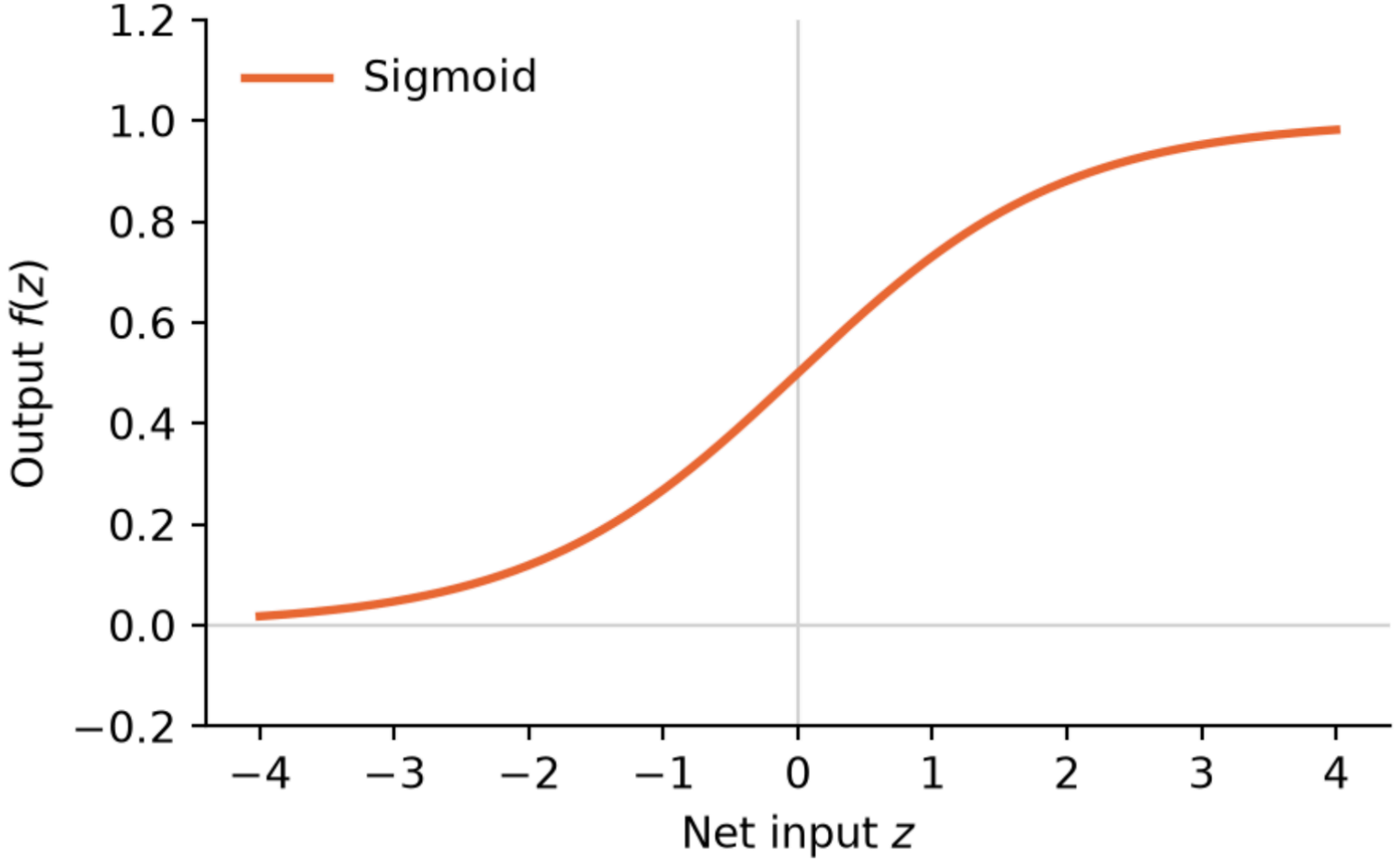
Activation function

$$f(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{otherwise'} \end{cases}$$



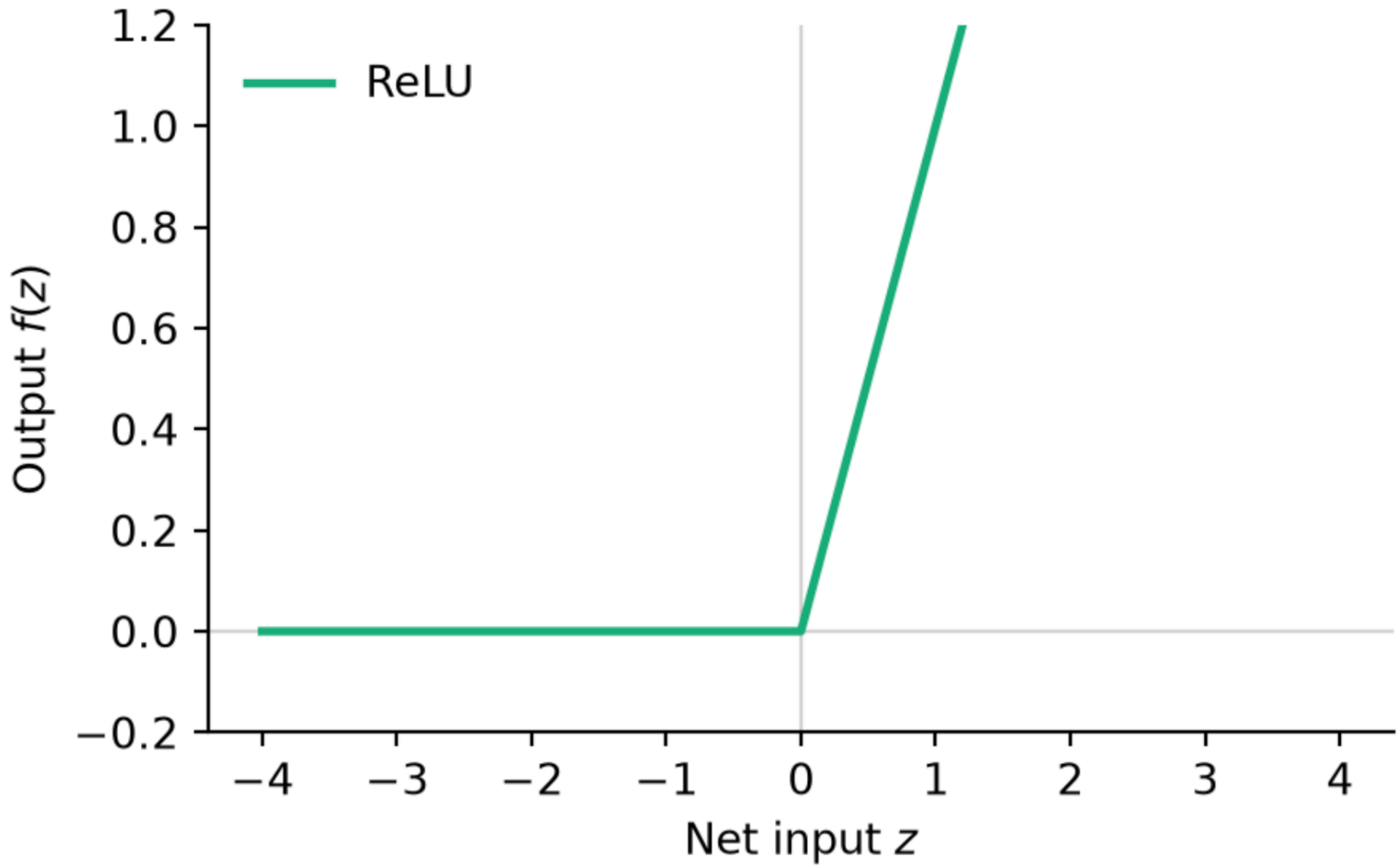
Activation function

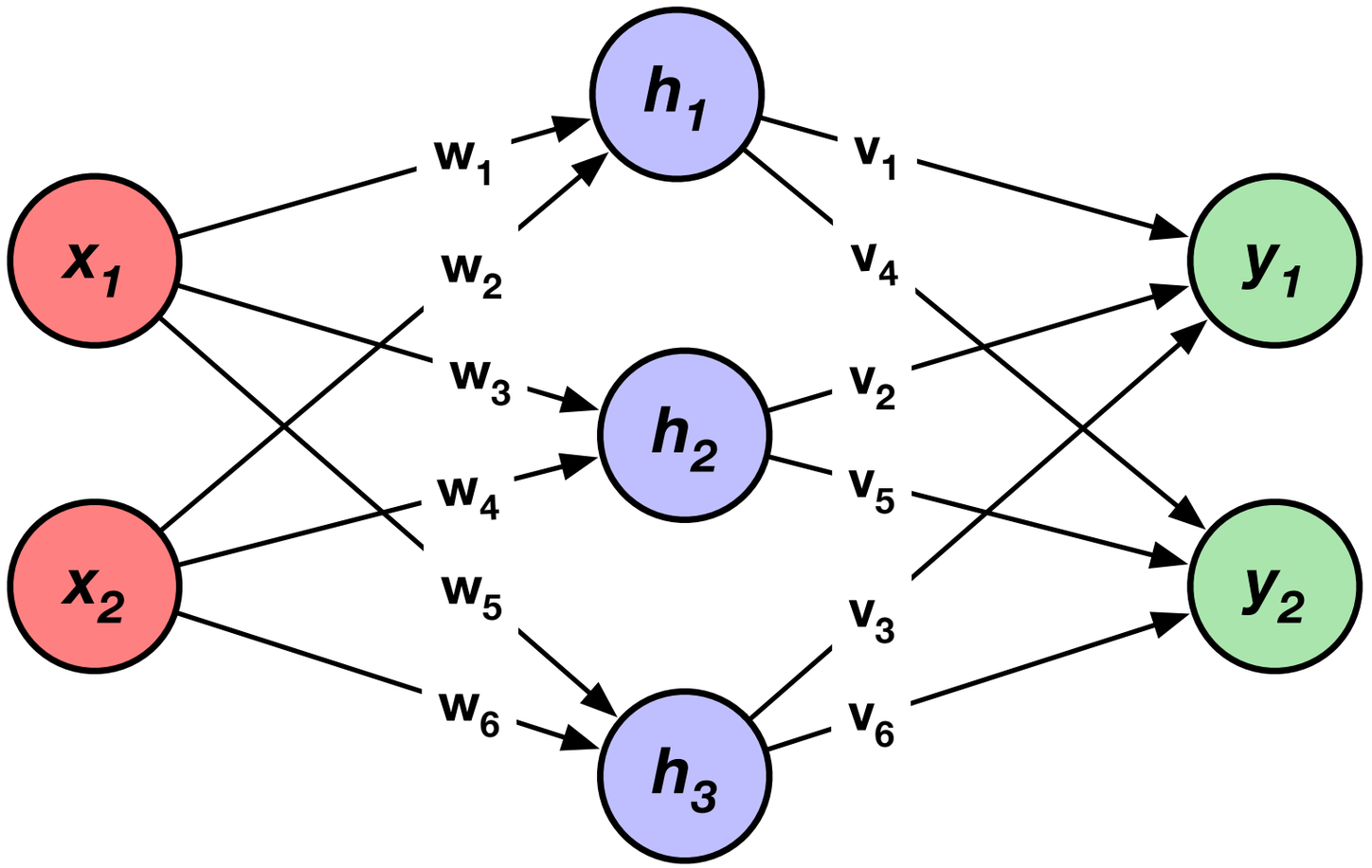
$$f(z) = \frac{1}{1 + e^{-z}}$$

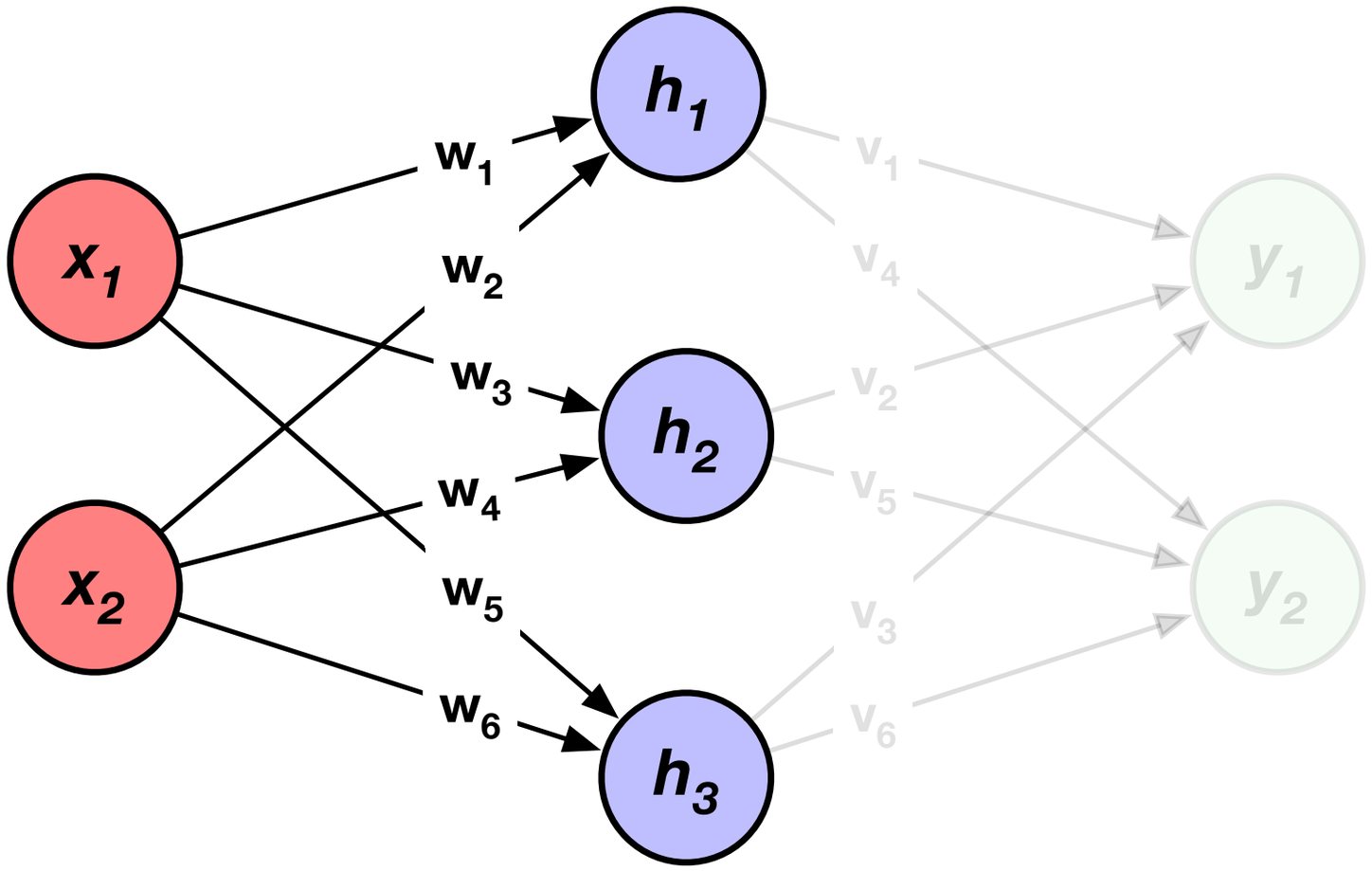


Activation function

$$f(z) = \max(0, z) = \begin{cases} z, & \text{if } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

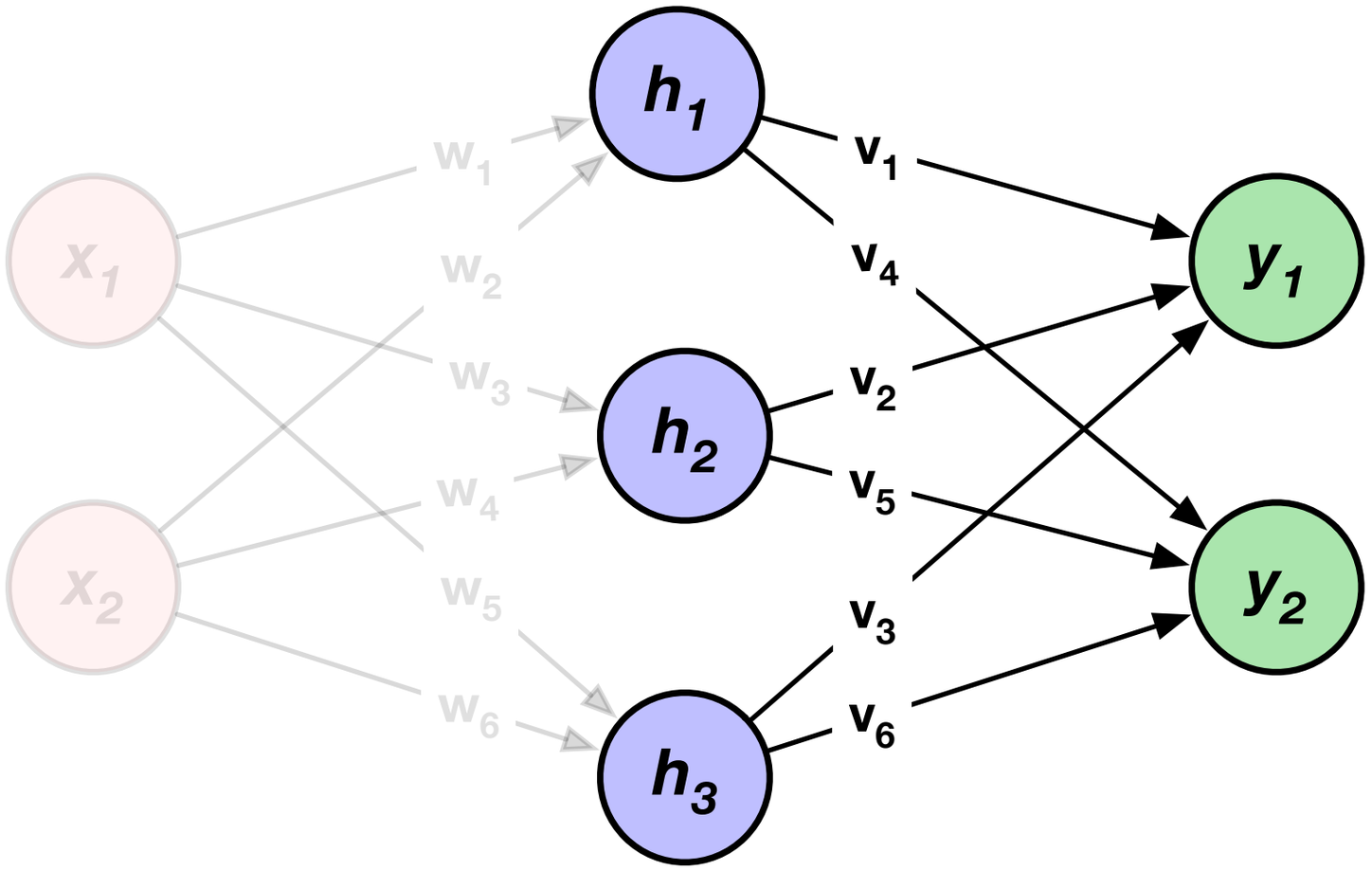



h_1
w_1
x_1
w_2
w_3
h_2
w_4
x_2
w_5
w_6
h_3
v_1
v_4
y_1
v_2
v_5
y_2
v_3
v_6



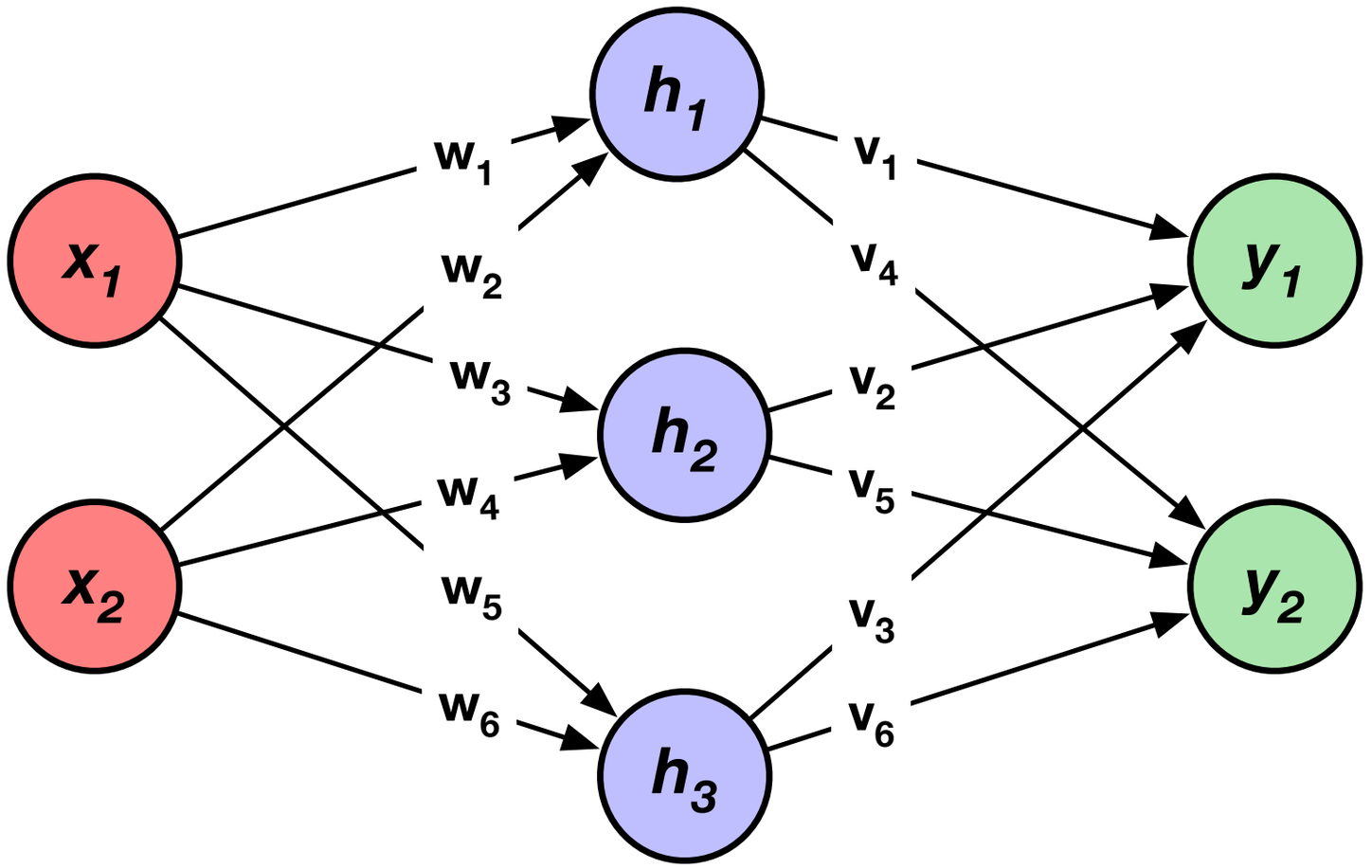
$$f\left(\begin{array}{c}\boldsymbol{X}\\ \begin{array}{|c|c|}\hline x_1 & x_2 \\ \hline\end{array}\end{array} \times \begin{array}{c}\boldsymbol{W}\\ \begin{array}{|c|c|c|}\hline w_1 & w_3 & w_5 \\ \hline w_2 & w_4 & w_6 \\ \hline\end{array}\end{array} - \begin{array}{c}\boldsymbol{\theta}\\ \begin{array}{|c|c|c|}\hline \theta_1 & \theta_2 & \theta_3 \\ \hline\end{array}\end{array}\right) = \begin{array}{c}\boldsymbol{H}\\ \begin{array}{|c|c|c|}\hline h_1 & h_2 & h_3 \\ \hline\end{array}\end{array}$$

$$f(XW - \theta) = H$$

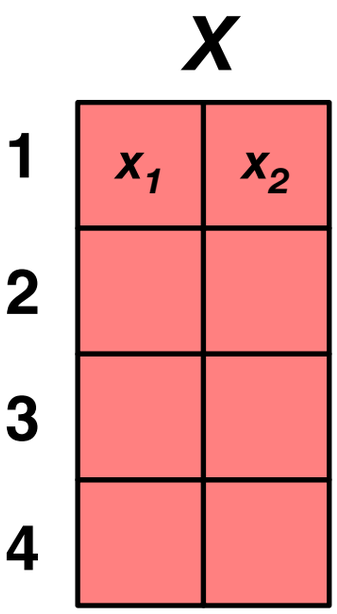


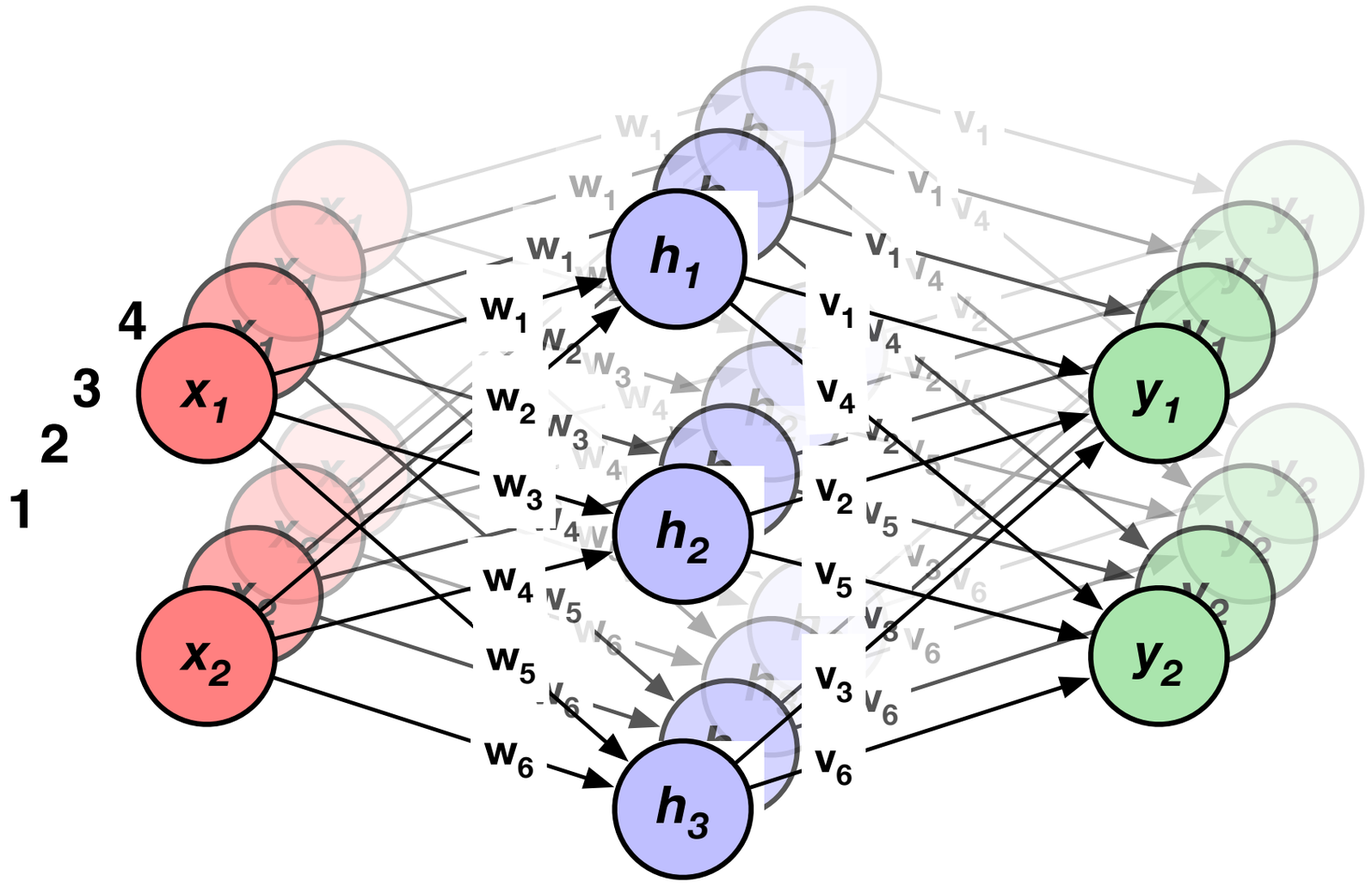
$$f\left(\begin{array}{|c|c|c|} \hline h_1 & h_2 & h_3 \\ \hline \end{array} \times \begin{array}{|c|c|} \hline v_1 & v_4 \\ \hline v_2 & v_5 \\ \hline v_3 & v_6 \\ \hline \end{array} - \begin{array}{|c|c|c|} \hline \theta_1 & \theta_2 & \theta_3 \\ \hline \end{array} \right) = \begin{array}{|c|c|} \hline y_1 & y_2 \\ \hline \end{array}$$

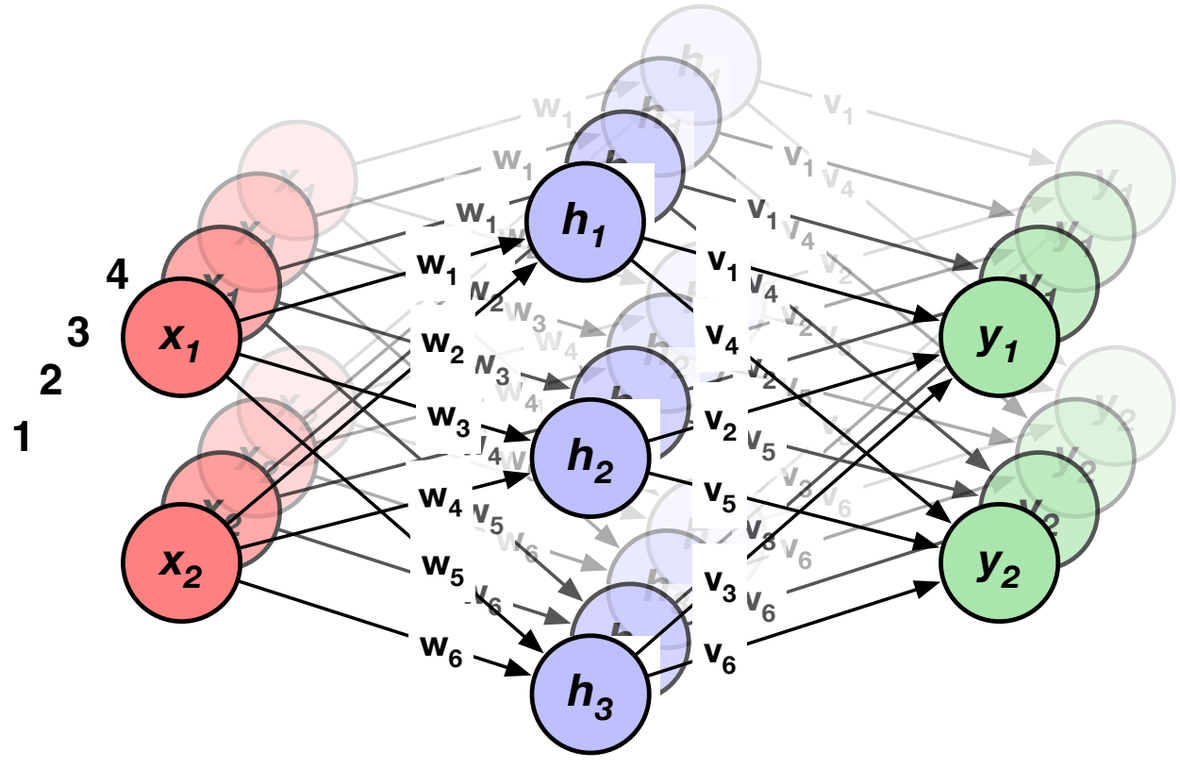
$$f(HV - \theta) = Y$$



$$f(XW - \theta) = H \qquad f(HV - \theta) = Y$$


X
1
2
3
4
x_1
x_2

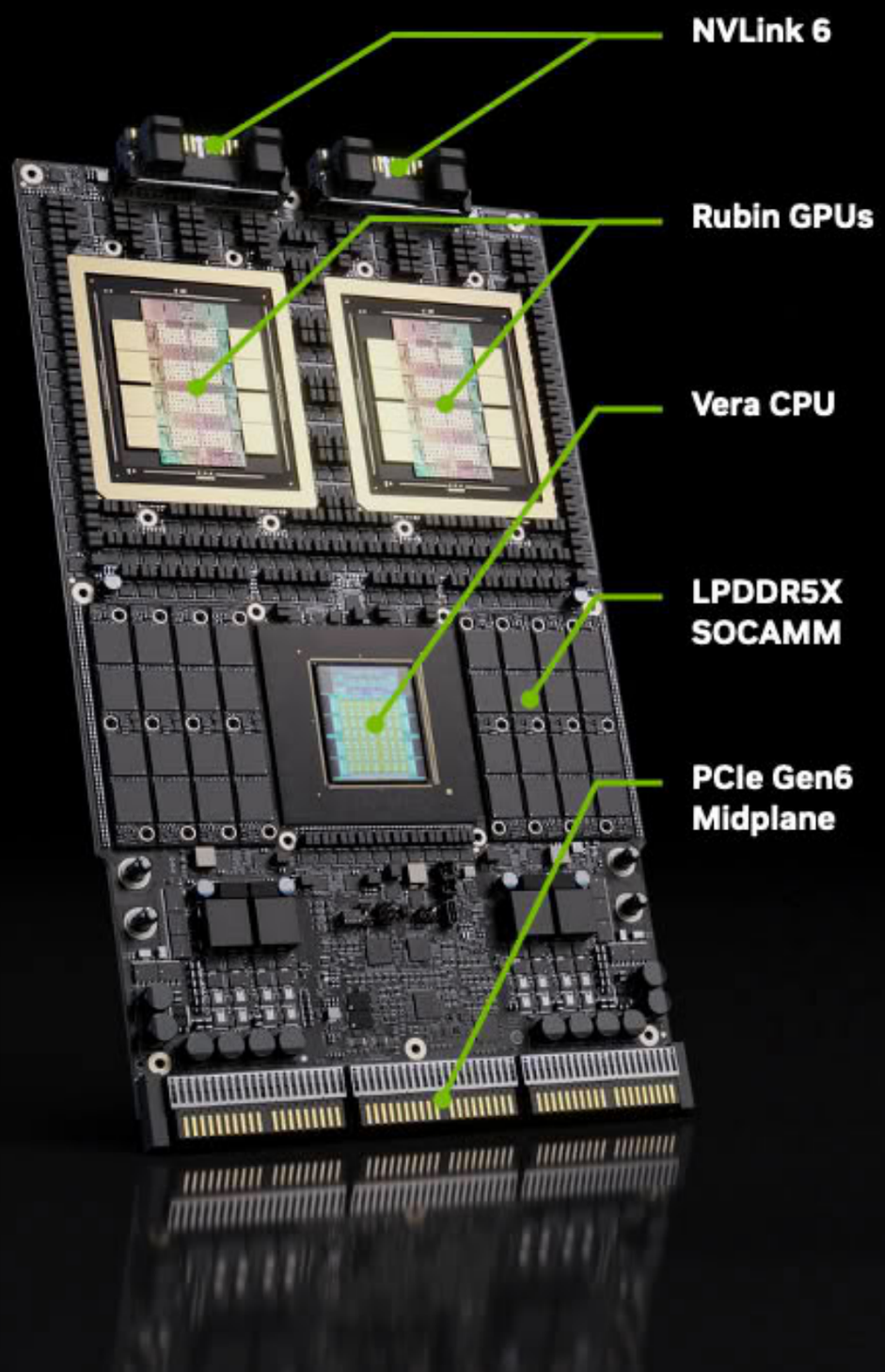

4
3
2
1
x_1
x_2
h_1
h_2
h_3
y_1
y_2
w_1
w_2
w_3
w_4
w_5
w_6
v_1
v_2
v_3
v_4
v_5
v_6



$$f\left(\begin{array}{|c|c|}\hline x_1 & x_2 \\ \hline & \\ \hline & \\ \hline & \\ \hline\end{array} \times \begin{array}{|c|c|c|}\hline w_1 & w_3 & w_5 \\ \hline w_2 & w_4 & w_6 \\ \hline\end{array} - \begin{array}{|c|c|c|}\hline \theta_1 & \theta_2 & \theta_3 \\ \hline\end{array}\right) = \begin{array}{|c|c|c|}\hline h_1 & h_2 & h_3 \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline\end{array}$$

$$f\left(\begin{array}{|c|c|c|}\hline h_1 & h_2 & h_3 \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline\end{array} \times \begin{array}{|c|c|}\hline v_1 & v_4 \\ \hline v_2 & v_5 \\ \hline v_3 & v_6 \\ \hline\end{array} - \begin{array}{|c|c|c|}\hline \theta_1 & \theta_2 & \theta_3 \\ \hline\end{array}\right) = \begin{array}{|c|c|}\hline y_1 & y_2 \\ \hline & \\ \hline & \\ \hline & \\ \hline\end{array}$$

100 PFLOPS = One hundred million billion floating point operations / sec



NVIDIA Vera Rubin Superchip

Processor for Gigascale AI Factories

- 100 PFLOPS AI Compute
- 88 NVIDIA Custom Olympus CPU Cores
- 2 TB Fast Memory

